

Veterinary Package Insert

TIMECOX 200 SOL TILMICOSIN 200G/L (20% Tilmicosin)

PRODUCT DESCRIPTION

Timecox 200 Sol Tilmicosin 200g/L is a light brown to brown colour liquid, which contains 20% tilmicosin, in 1 L bottle.

PHARMACODYNAMICS & PHARMACOKINETICS

Tilmicosin is a mainly bactericidal semi-synthetic antibiotic of the macrolide group. It is believed to affect bacterial protein synthesis.

Tilmicosin has a wide spectrum of activity against Gram-positive organisms and is particularly active against *Pasteurella*, *Actinobacillus* (*Haemophilus*) and *Mycoplasma* organisms of bovine, porcine and avian origin. Tilmicosin has some activity against certain Gram-negative micro-organisms. Cross resistance between tilmicosin and other macrolide antibiotics has been observed. Macrolides inhibit protein synthesis by reversibly binding to the 50S ribosomal subunit. Bacterial growth is inhibited by induction of the separation of peptidyl transfer RNA from the ribosome during the elongation phase.

Ribosomal methylase, encoded by the *erm* gene, can precipitate resistance to macrolides by alteration of the ribosomal binding site.

The gene that encodes for an efflux mechanism, *mef*, also brings about a moderate degree of resistance.

Resistance is also brought about by an efflux pump that actively rids the cells of the macrolide. This efflux pump is chromosomally mediated by genes referred to as *acrAB* genes. Resistance of *Pseudomonas* species and other Gram-negative bacteria, enterococci and staphylococci may be precipitated by chromosomally controlled alteration of permeability or uptake of the drug.

When administered orally to chickens, turkeys and pigs with drinking water, tilmicosin is absorbed and moves rapidly out of the serum into areas of low pH. This results in very low serum concentrations, but detectable levels of tilmicosin are found in lung tissue as early as 6 hours after starting the treatment. In chicken or turkeys, tilmicosin is also detected in pooled air sac tissue as early as 6 hours after starting the treatment. It is also known that tilmicosin is concentrated in alveolar macrophages of swine.

INDICATION

Chickens (broilers and pullets):

For the treatment and prevention of respiratory infections in poultry flocks associated with *Mycoplasma gallisepticum* and *Mycoplasma synoviae* when the disease has been diagnosed at the herd level.

Turkeys:

For the treatment and prevention of respiratory infections in turkey flocks associated with *Mycoplasma gallisepticum* and *Mycoplasma synoviae* when the disease has been diagnosed at the herd level.

Pigs:

For the treatment and prevention of respiratory infections associated with *Mycoplasma hyopneumoniae*, *Pasteurella multocida* and *Actinobacillus pleuropneumoniae* when the disease has been diagnosed at the herd level.

RECOMMENDED DOSAGE AND ADMINISTRATION

It is to be given orally. The product must be diluted in drinking water or milk replacer before administration. The recommended dosage for is as following:

Chickens (Broilers and Pullets):

15-20 mg tilmicosin per kg body weight for 3 days, i.e. 7.5-10 ml of product for 100 kg body weight corresponding to 37.5 ml of product per 100 litres of drinking water for 3 days.

Turkeys:

10-27 mg tilmicosin per kg body weight for 3 days, i.e. 5-13.5 ml of product for 100 kg body weight corresponding to 37.5 ml of product per 100 litres of drinking water for 3 days.

Pigs:

15-20 mg tilmicosin per kg body weight for 5 days, i.e. 7.5-10 ml of product for 100 kg body weight corresponding to 100 ml of product per 100 litres of drinking water for 5 days.

CONTRAINDICATIONS

Do not use in case of hypersensitivity to the active substance or in cases of known resistance to tilmicosin. Do not use in horses.

WARNINGS & PRECAUTIONS.

Special precautions for use in animals:

Inappropriate use of the product may increase the prevalence of bacteria resistant to tilmicosin and may decrease the effectiveness of treatment with tilmicosin related substances. Use of the product should be based on susceptibility testing and take into account official and local antimicrobial policies.

Do not allow horses or other equines access to drinking water containing tilmicosin

Special precautions to be taken by the person administering the veterinary medicinal product to animals:

People with known hypersensitivity to tilmicosin should avoid contact with the product. The veterinary medicinal product may cause irritation or sensitisation by skin contact.

Avoid skin and ocular contact. Wear protective gloves and protective clothes when handling the veterinary medicinal product.

In case of contact with skin or eyes, rinse abundantly with fresh water. If irritation persists and in case of incidental ingestion, seek immediately medical advice or call a poison center (dangers linked to disturbances in cardiac conduction).

Wash hands after use.

INTERACTIONS WITH OTHER MEDICATIONS

Cross resistance between tilmicosin and other macrolide antibiotics and lincosamides has been observed.

Tilmicosin may lessen the antibacterial activity of β -lactam antibiotics

Do not use simultaneously with bacteriostatic antimicrobial agents.

PREGNANCY AND LACTATION

The safety of the product has not been established during pregnancy and lactation. Use only in accordance with risk/benefit assessment by the responsible veterinarian.

ADVERSE EFFECTS

None known.

OVERDOSE & TREATMENT

Pigs drink less water when a dose of 300 to 400 mg/liter (1.5 to 2 times the recommended dose) is administered. Although this will result in less intake of tilmicosin, it might lead to dehydration of the animals. Replace with untreated water when needed.

No symptoms were seen in poultry treated at 375 mg/liter during 5 days. A dose of 75 mg/liter during 10 days resulted in less consistent faeces.

No symptoms of overdose were noticed in turkeys treated at 375 mg/liter of drinking water during 3 days. No symptoms were noticed at 75 mg/liter during 6 days.

Except for a slight decrease in milk intake, no symptoms of overdose were seen in calves treated

at 5 times the recommended dose or during twice the recommended treatment period.

WITHDRAWAL PERIOD

Chickens (Broilers and Pullets):

12 days.

Turkey:

19 days.

Pigs:

14 days.

Not authorised for use in laying birds producing eggs for human consumption.

MAXIMUM RESIDUAL LIMIT (MRL)

Substance	Food	Maximum Residue Limits (MRLs) in $\mu\text{g/kg}$
Tilmicosin	Muscle, fat (poultry and pig)	100
	Kidney (pig)	1000
	Liver (pig)	1500

STORAGE CONDITION

Store below 30°C. Protect from direct sunlight.

SHELF LIFE

2 years.

After 1st opening, reconstitution or dilution, use within 24 hours.

PACKING

1L

LIM SENG PHARMACHEM SDN. BHD.

(product registration holder)
2692, Tingkat Perusahaan Enam,
Prai Industrial Estate MK1,
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THYE PHARMA SDN. BHD.

(manufacturer)
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